Cocoercive flow approximation and odd Mann iterations

Abstract

We give two uniform-in-time comparisons between a cocoercive flow and its forward Euler discretization. When the operator has a zero and the steps are bounded strictly below the stability threshold, the errors are controlled by the remaining residual energy. Square-summability of the steps is unnecessary. The comparisons yield mutual almost-orbit approximation and strong convergence of odd nonlinear Mann iterations when the relaxation sum diverges and the upper limit is less than one. A separate finite-weight coupling treats eventually nondecreasing relaxations under the usual divergent-series condition, including schedules approaching one. It gives a deterministic Cauchy bound in terms of the remaining norm energy. These are restricted cases of a recent odd-map question; arbitrary oscillating endpoint-approaching schedules remain untreated. A nonlinear example lies outside strict averagedness, and a projection example shows that the limit need not be the closest fixed point to the initial state.

Keywords: cocoercivity; Euler iteration; almost-orbits; strong convergence

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1 Introduction

Let T be an odd nonexpansive map on a real Hilbert space. Consider the unchanged Mann iteration

$$x_{n+1} = (1-\lambda_n)x_n + \lambda_n T x_n, \qquad 0 \le \lambda_n \le 1. \tag{1}$$

Bartz, Bauschke and Gao [1] prove strong convergence under $\sum_n \lambda_n(1-\lambda_n) = \infty$ when T is linear. Their accepted version, Section 3, asks about replacing linearity by oddness. Strong convergence for powers of one odd asymptotically regular nonexpansive map is classical [2, 3]; changing relaxation parameters prevents a direct application of that fixed-map statement.

Our first route proves the odd nonlinear conclusion when

$$\sum_n \lambda_n = \infty, \qquad \limsup_{n\to\infty} \lambda_n < 1. \tag{2}$$

No positive lower bound, finite variation, or convergence of the parameters is required. Under the upper bound in (2), divergence of the sum is equivalent to the usual divergent-series condition.

A second route proves strong convergence when the parameters are eventually nondecreasing and $\sum_n \lambda_n(1-\lambda_n) = \infty$, without a uniform gap below one. In particular it

covers $\lambda_n = 1 - 1/(n+2)$. The proof gives an explicit norm-Cauchy estimate by a finite-weight comparison of two orbit indices. Bernoulli, Markov-chain and optimal-transport comparisons for nonlinear Mann iterations have established ancestry [11–14]. The use here combines oddness with the summable Mann energy; the coupling device itself is not claimed as new. Arbitrary nonmonotone schedules approaching the upper endpoint are not settled.

The argument uses energy-controlled comparisons for a general cocoercive operator with a zero. Euler stability and its step threshold are established facts; see, for example, the contractivity framework of Dahlquist and Jeltsch [4]. Continuous–discrete comparison and almost-orbit transfer also have substantial ancestry [5–8]. The general comparison in [7] uses square-summable steps, whereas its potential-operator section removes that restriction. Its zero-containing Euler estimate already supplies the residual summability used below. The estimates here retain that residual inside the comparison error, in both directions, for arbitrary cocoercive operators with a zero. The forward–backward extension [8] likewise uses square-summable steps in its general infinite-horizon comparison.

The contribution of this note is the pair of explicit energy-tail comparisons in the zero-containing class and the strong-convergence conclusions for the two stated schedule classes. The classical stability, dissipation, almost-orbit principle, odd-flow argument and probabilistic-comparison ancestry are retained and credited. All proofs needed for the conclusions below are included.

2 Two energy-controlled comparisons

Throughout this section, $\mathcal{H}$ is a real Hilbert space and $A : \mathcal{H} \to \mathcal{H}$ is β-cocoercive:

$$\langle Ax - Ay, x - y\rangle \geq \beta\|Ax - Ay\|^2, \qquad x, y \in \mathcal{H}, \quad \beta > 0. \tag{3}$$

Assume $\operatorname{zer} A \neq \varnothing$ and fix $p \in \operatorname{zer} A$. The operator is monotone and $1/\beta$-Lipschitz. Thus

$$u'(t) = -Au(t), \qquad u(0) = x \tag{4}$$

has a unique global continuously differentiable solution, denoted by $S(t)x$. Monotonicity makes $S(t)$ nonexpansive.

Let

$$x_{n+1} = x_n - h_n Ax_n, \qquad 0 < h_n \leq \bar{h} < 2\beta, \qquad \sum_{n=0}^{\infty} h_n = \infty. \tag{5}$$

Zero steps can always be deleted. Put

$$t_n = \sum_{j=0}^{n-1} h_j, \qquad X(t) = x_n - (t - t_n)Ax_n \quad (t_n \leq t \leq t_{n+1}). \tag{6}$$

Lemma 1. *The limits*

$$E_\infty = \lim_n \|x_n - p\|^2, \qquad F_\infty = \lim_{t\to\infty} \|u(t) - p\|^2$$

exist. For every N and $s \geq 0$,

$$\sum_{n=N}^{\infty} h_n(2\beta - h_n)\|Ax_n\|^2 \leq \|x_N - p\|^2 - E_\infty, \tag{7}$$

$$2\beta \int_s^\infty \|Au(t)\|^2\, dt \leq \|u(s) - p\|^2 - F_\infty. \tag{8}$$

Moreover, $\|Ax_n\|$ and $\|Au(t)\|$ are nonincreasing and tend to zero.

Proof. Expanding one Euler step and applying (3) with $y = p$ gives

$$\|x_{n+1} - p\|^2 \leq \|x_n - p\|^2 - h_n(2\beta - h_n)\|Ax_n\|^2.$$

The analogous flow identity is $\frac{d}{dt}\|u-p\|^2 = -2\langle Au, u-p\rangle \leq -2\beta\|Au\|^2$. Summation and integration prove the two tail bounds.

For $a = Ax_n$ and $d = Ax_{n+1} - Ax_n$, cocoercivity gives $\beta\|d\|^2 \leq -h_n\langle d, a\rangle$. Consequently

$$\|Ax_{n+1}\|^2 \leq \|Ax_n\|^2 - \left(\frac{2\beta}{h_n} - 1\right)\|d\|^2.$$

The discrete residual therefore decreases, and its limit is zero by (7), $\bar{h} < 2\beta$, and $\sum_n h_n = \infty$. Nonexpansivity of S makes $\|u(t+\varepsilon) - u(t)\|/\varepsilon$ nonincreasing in t for each $\varepsilon > 0$. Letting $\varepsilon \downarrow 0$ gives monotonicity of $\|Au(t)\|$; its limit is zero by (8). ∎

Theorem 1. *Under (3)–(5), the following estimates hold.*
(i) *For every N and every $t \geq t_N$,*

$$\|X(t) - S(t - t_N)x_N\|^2 \leq \frac{1}{6\beta}\sum_{n=N}^{\infty} h_n^3\|Ax_n\|^2 \tag{9}$$

$$\leq \frac{\bar{h}^2}{6\beta(2\beta - \bar{h})}\big(\|x_N - p\|^2 - E_\infty\big).$$

(ii) *Fix a flow orbit u, a time $s \geq 0$, and any future grid with steps in $(0, \bar{h}]$, writing its partial sums again as t_n. Starting at $v_0 = u(s)$, set $v_{n+1} = v_n - h_n Av_n$. Then*

$$\|v_n - u(s + t_n)\|^2 \leq \frac{\bar{h}^2}{2(2\beta - \bar{h})}\int_s^{s+t_n} \|Au(t)\|^2\, dt \tag{10}$$

$$\leq \frac{\bar{h}^2}{4\beta(2\beta - \bar{h})}\big(\|u(s) - p\|^2 - F_\infty\big).$$

The reverse estimate does not require divergence of the future step sum. Both bounds vanish at late starting times, uniformly over all future horizons; the second is also uniform over the permitted future grids.

Proof. (i) Fix N and write $u(t) = S(t - t_N)x_N$ for $t \geq t_N$. On the nth interpolation interval put $r = t - t_n$, $e = X(t) - u(t)$ and $D = Ax_n - Au(t)$. Since $e = x_n - u - rAx_n$, cocoercivity and completion of a square give

$$\begin{aligned}\frac{d}{dt}\|e\|^2 &= -2\langle D, x_n - u\rangle + 2r\langle D, Ax_n\rangle \\ &\leq -2\beta\|D\|^2 + 2r\|D\|\|Ax_n\| \\ &\leq \frac{r^2}{2\beta}\|Ax_n\|^2.\end{aligned}$$

Integration over a full step gives $h_n^3\|Ax_n\|^2/(6\beta)$. The initial error is zero, and a partially traversed final step only reduces the bound. This proves the first inequality in (9). The second follows from (7) and $h_n^3 \leq \bar{h}^2 h_n(2\beta - h_n)/(2\beta - \bar{h})$.

(ii) Consider one step of length h, with Euler state v and flow starting time r_0. Define

$$e = v - u(r_0), \qquad D(r) = Av - Au(r_0 + r), \qquad w(r) = u(r_0) - u(r_0 + r).$$

The next error is exactly $e_+ = e - \int_0^h D(r)\,dr$. Because $e = v - u(r_0 + r) - w(r)$, the integral Cauchy–Schwarz inequality and cocoercivity imply

$$\begin{aligned}\|e_+\|^2 - \|e\|^2 &\leq -(2\beta - h)\int_0^h \|D(r)\|^2\,dr + 2\int_0^h \langle D(r), w(r)\rangle\,dr \\ &\leq \frac{1}{2\beta - h}\int_0^h \|w(r)\|^2\,dr.\end{aligned}$$

Also $w(r) = \int_0^r Au(r_0 + a)\,da$, so

$$\int_0^h \|w(r)\|^2\,dr \leq \int_0^h r\int_0^r \|Au(r_0 + a)\|^2\,da\,dr \leq \frac{h^2}{2}\int_0^h \|Au(r_0 + a)\|^2\,da.$$

Sum the resulting one-step bound from the zero initial error. Since $h^2/(2\beta - h) \leq \bar{h}^2/(2\beta - \bar{h})$, this proves (10); its energy form follows from (8). ∎

3 The discrete clock and almost-orbits

We spell out the time convention because the approximation above is uniform on an infinite future interval. Define

$$\nu(t) = \max\{n : t_n \leq t\}, \qquad F_n = I - h_n A,$$

and

$$P_{m,n} = F_{m-1}\cdots F_n \quad (m > n), \qquad P_{n,n} = I.$$

The nonexpansive evolution systems are

$$U_E(t, s) = P_{\nu(t),\nu(s)}, \qquad U_S(t, s) = S(t - s), \qquad t \geq s \geq 0. \tag{11}$$

A function ϕ is an almost-orbit of U when

$$\lim_{t\to\infty} \sup_{h\geq 0} \|\phi(t+h) - U(t+h,t)\phi(t)\| = 0. \tag{12}$$

The terminology and the general transfer principle are from the classical evolution-system framework; see [6].

Theorem 2. *Every orbit of U_E is an almost-orbit of U_S, and every orbit of U_S is an almost-orbit of U_E. Here an orbit may start at any fixed time with any initial state. In particular, strong convergence of all orbits of either system implies strong convergence of all orbits of the other.*

Proof. First take the Euler orbit $\phi_E(t) = x_{\nu(t)}$. For $h \geq 0$ put $n = \nu(t)$ and $m = \nu(t+h)$. Then

$$(t_m - t_n) - h = (t - t_n) - (t + h - t_m), \qquad |(t_m - t_n) - h| \leq \bar{h}. \tag{13}$$

The forward comparison bounds $\|x_m - S(t_m - t_n)x_n\|$ by the square root of an Euler energy tail. The remaining clock error is at most $\bar{h}\|Ax_n\|$, since the residual along the flow starting from x_n is nonincreasing. Both bounds tend to zero uniformly in h, proving (12).

Conversely, start an Euler sequence at $u(t)$ with steps $h_n, \ldots, h_{m-1}$. The reverse comparison bounds $\|P_{m,n}u(t) - u(t + t_m - t_n)\|$ by the square root of a flow energy tail beginning at t, independent of m and the grid. By (13), its difference from $u(t+h)$ is at most $\bar{h}\|Au(t)\|$. This proves the reverse almost-orbit statement. Shifting the initial index gives the same estimates for an Euler orbit started at any time; the flow argument likewise permits any fixed starting time.

For completeness, let ϕ be an almost-orbit of a system whose every orbit converges strongly. Choose a late time t so that the supremum in (12) is less than $\varepsilon/3$. The orbit $h \mapsto U(t+h,t)\phi(t)$ is eventually Cauchy. The triangle inequality makes ϕ eventually ε-Cauchy. Completeness proves the last assertion. ∎

Remark. The all-orbit statement includes all starting times, not only all initial states at index zero. A finite Euler prefix need not be surjective. No equality between the limit points of two differently initialized evolutions is asserted. The presence of a zero and the strict bound $\bar{h} < 2\beta$ are retained throughout; these results do not replace a zero-free comparison theorem.

4 Strong convergence for odd Mann iterations

We give the classical odd-flow argument in the form needed here; its fixed-map and monotone-flow ancestry is discussed in [2, 3, 9].

Lemma 2. *If A is odd and cocoercive, every solution of (4) converges strongly to a zero of A.*

Proof. Oddness gives $A0 = 0$ and, by uniqueness, $S(t)(-x) = -S(t)x$. Write $r = \lim_{t\to\infty} \|u(t)\|$, which exists by Lemma 1. For fixed $s, t \geq 0$ and $k \geq 0$, oddness and nonexpansivity give

$$\|u(s+k) + u(t+k)\| \leq \|u(s) + u(t)\|.$$

Furthermore,

$$\|u(s+k) - u(t+k)\| \leq |s-t|\,\|Au(k+\min\{s,t\})\| \longrightarrow 0.$$

The individual vector norms tend to r, and the vectors' difference vanishes. Thus $\|u(s+k) + u(t+k)\| \to 2r$ and $\|u(s) + u(t)\| \geq 2r$. The parallelogram identity yields

$$\|u(s) - u(t)\|^2 \leq 2\big(\|u(s)\|^2 - r^2\big) + 2\big(\|u(t)\|^2 - r^2\big).$$

This is a strong Cauchy estimate. The limit is a zero because $Au(t) \to 0$ and A is continuous. ∎

Theorem 3. *Let $T : \mathcal{H} \to \mathcal{H}$ be odd and nonexpansive, and let $x_0 \in \mathcal{H}$. If $\lambda_n \in [0,1]$ satisfies* (2)*, then* (1) *converges strongly to some point of* $\operatorname{Fix} T$.

Proof. Set $A = I - T$. Expanding a square and using nonexpansivity shows

$$\|Ax - Ay\|^2 \leq 2\langle Ax - Ay, x - y\rangle.$$

Thus A is 1/2-cocoercive, is odd, and has the zero 0. Discard a finite prefix so that $\lambda_n \leq q < 1$ and delete zero steps. The iteration is exactly (5) with $h_n = \lambda_n$. Lemma 2 gives strong convergence of every flow orbit. The forward bound (9) makes the Euler interpolation uniformly close, from any sufficiently late starting point onward, to one such orbit. The Cauchy argument in Theorem 2 proves strong convergence. Lemma 1 and continuity identify the limit as a zero of A. ∎

5 Nondecreasing relaxations reaching the endpoint

This section does not use the strict upper step bound from the flow comparison. We work directly with (1). The auxiliary weights below are a proof device; the iteration and its conclusion remain deterministic.

Theorem 4. *Let $T : \mathcal{H} \to \mathcal{H}$ be odd and nonexpansive. Suppose $\lambda_n \in [0,1]$ is eventually nondecreasing and*

$$\sum_{n=0}^{\infty} \lambda_n(1-\lambda_n) = \infty. \tag{14}$$

For arbitrary x_0, iteration (1) *converges strongly to some fixed point of T. More precisely, put*

$$r_n = \|Tx_n - x_n\|, \qquad \rho = \lim_n \|x_n\|, \qquad E_n = \sum_{k=n}^{\infty} \lambda_k(1-\lambda_k)r_k^2.$$

For all $m \geq n$ on a positive nondecreasing tail,

$$\|x_n + x_m\|^2 \geq 4\rho^2 - E_n, \tag{15}$$
$$\|x_n - x_m\|^2 \leq 4(\|x_n\|^2 - \rho^2) + E_n \leq 5(\|x_n\|^2 - \rho^2). \tag{16}$$

The same estimates and convergence hold if (14) *is replaced by eventual positivity of the nondecreasing parameters and the assumption $r_n \to 0$.*

Proof. *Energy and residuals.* Oddness gives $T0 = 0$. The convex-combination identity yields

$$\|x_{k+1}\|^2 \leq \|x_k\|^2 - e_k, \qquad e_k = \lambda_k(1 - \lambda_k)r_k^2. \tag{17}$$

Consequently ρ exists and

$$E_n \leq \|x_n\|^2 - \rho^2. \tag{18}$$

For any schedule in $[0, 1]$, nonexpansivity also gives

$$r_{k+1} \leq \|Tx_{k+1} - Tx_k\| + \|Tx_k - x_{k+1}\| \leq \lambda_k r_k + (1 - \lambda_k)r_k = r_k.$$

Thus (14) implies $r_k \to 0$ by (17). It also ensures a strictly positive parameter on the monotone tail, after which every parameter is positive. It remains to prove the estimates under these latter two premises.

One-step finite-weight comparison. Fix $m \geq n$ on the positive nondecreasing tail and let $d = m - n$. For $k \geq n$ and $k \leq j \leq k + d$, define

$$S_a = (1 - a)I + aT \quad (0 \leq a \leq 1), \qquad q = \frac{\lambda_k}{\lambda_j} \in (0, 1].$$

Each S_a is odd and nonexpansive. Moreover, $(1 - q)x_j + qx_{j+1} = S_{\lambda_k}x_j$. The two-point variance identity therefore gives

$$\begin{aligned}
&(1 - q)\|x_{k+1} + x_j\|^2 + q\|x_{k+1} + x_{j+1}\|^2 \\
&\quad = \|S_{\lambda_k}x_k + S_{\lambda_k}x_j\|^2 + q(1 - q)\|x_{j+1} - x_j\|^2 \\
&\quad \leq \|x_k + x_j\|^2 + \lambda_k(\lambda_j - \lambda_k)r_j^2 \leq \|x_k + x_j\|^2 + e_k.
\end{aligned} \tag{19}$$

The last inequality uses $r_j \leq r_k$ and $0 \leq \lambda_j - \lambda_k \leq 1 - \lambda_k$.

Start with unit mass $p_n(m) = 1$. At time k, move mass at each index j to $j + 1$ with proportion λ_k/λ_j, and retain the remaining proportion at j. This defines finite nonnegative weights p_{k+1} of total mass one. Their support is contained in $\{k+1, \ldots, k+d+1\}$: any mass at $j = k$ moves with proportion one. Thus

$$A_k = \sum_{j=k}^{k+d} p_k(j)\|x_k + x_j\|^2$$

is well defined for every finite $k \geq n$. Summing (19) with weights $p_k(j)$ gives $A_{k+1} \leq A_k + e_k$, and hence

$$A_N \leq \|x_n + x_m\|^2 + \sum_{k=n}^{N-1} e_k, \qquad N \geq n. \tag{20}$$

Limit and all-future bound. Keep n, m, and therefore d, fixed while $N \to \infty$. For every $N \leq j \leq N + d$,

$$\|x_j - x_N\| \leq d\, r_N.$$

All these vectors have norm at most $\|x_n\|$. Consequently

$$\left|A_N - 4\|x_N\|^2\right| \leq 4\|x_n\|\, d\, r_N \longrightarrow 0.$$

This estimate holds over the whole moving support, so it does not require convergence of the weights. Passing to the limit in (20) proves (15). The parallelogram identity, norm monotonicity and (18) then give

$$\|x_n - x_m\|^2 \leq 2(\|x_n\|^2 - \rho^2) + 2(\|x_m\|^2 - \rho^2) + E_n,$$

and thus (16). Although the auxiliary limit was taken for a fixed pair, the final bound has no dependence on d. It therefore holds for every $m \geq n$ with the same starting index. Its right-hand side tends to zero, proving that (x_n) is Cauchy. Continuity of T and $r_n \to 0$ identify its norm limit as a fixed point. ∎

For the harmonic schedule $\lambda_n = 1 - 1/(n+2)$, the divergent series has terms $(n+1)/(n+2)^2$. Thus Theorem 4 genuinely permits approach to the endpoint. Under the weaker residual-decay formulation it even permits a tail equal to one, recovering the classical odd powers case. An eventually nonincreasing schedule satisfying (14) has a strict upper gap on a sufficiently late tail, so it is already covered by Theorem 3. No ordering assumption is needed there.

6 Scope examples and the remaining schedules

Example 1. The map in Theorem 3 need not be linear, a gradient, or strictly averaged. On $\ell^2(\mathbb{N};\mathbb{R})$ let $Re_j = e_{j+1}$ be the right shift, fix $c > 0$, and let $a_j \in (0, 1]$ tend to zero. Define

$$K = c(R - R^*), \qquad D(x)_j = a_j \tanh(x_j), \qquad B = K + D, \qquad T = 2(I + B)^{-1} - I.$$

The operator B is globally Lipschitz, odd, and monotone: K is skew-adjoint and D is the gradient of $\sum_j a_j \log\cosh(x_j)$, which is finite since $0 \leq \log\cosh t \leq t^2/2$. Continuity and full domain make B maximally monotone, so its resolvent is everywhere defined and firmly nonexpansive [10]. Thus T is odd and nonexpansive. Also

$$\langle Bx, x\rangle = \sum_j a_j x_j \tanh(x_j) > 0 \quad (x \neq 0),$$

so $\operatorname{Fix} T = \operatorname{zer} B = \{0\}$. If T were linear, its bijective resolvent $(I+T)/2$ would be linear, forcing B to be linear. This contradicts the hyperbolic tangent term. Its derivative at zero, $M = K + \operatorname{diag}(a_j)$, is not self-adjoint, so B is nonpotential. Indeed the coordinatewise remainder is $O(\|x\|^3)$ in ℓ^2. The derivative of T is $C = 2(I+M)^{-1} - I$; if T were a gradient, C would be self-adjoint, as would $M = 2(I+C)^{-1} - I$, a contradiction.

To see that T is not α-averaged for any $\alpha \in (0,1)$, suppose otherwise. Square expansion in the definition $T = (1-\alpha)I + \alpha Q$, with Q nonexpansive, gives

$$\|Tw\|^2 \le \|w\|^2 - \frac{1-\alpha}{\alpha}\|w - Tw\|^2.$$

For $z = e_j$, $j \ge 2$, put $w = z + Bz$. Then $Tw = z - Bz$, and the inequality would imply

$$a_j \tanh(1) = \langle z, Bz\rangle \ge \frac{1-\alpha}{\alpha}\|Bz\|^2 \ge \frac{1-\alpha}{\alpha}\, 2c^2,$$

which is impossible as $j \to \infty$. Theorems 3 and 4 therefore apply to a coupled nonlinear map outside strict averagedness. This example only records their scope; it is not a separate sharpness claim.

Example 2. In general the limit in Theorem 3 is not $P_{\operatorname{Fix} T} x_0$. Let $C_1 = \mathbb{R} \times \{0\}$ and $C_2 = \{(x,y) : |x-y| \le 1\}$. The map $T = P_{C_1} P_{C_2}$ is odd and nonexpansive, with $\operatorname{Fix} T = [-1,1] \times \{0\}$. For $\lambda_n = 1/2$ and $x_0 = (0,2)$, the first update is $x_1 = (1/4, 1)$, and induction gives

$$x_n = (1/4, 2^{1-n}), \qquad n \ge 1.$$

Indeed these points lie in C_2, so $Tx_n = (1/4, 0)$. The limit is $(1/4, 0)$, whereas $P_{\operatorname{Fix} T} x_0 = (0,0)$. This elementary example distinguishes the odd-map convergence conclusion from the projection formula in the linear theorem.

The estimates in Theorem 1 deteriorate as $\bar{h} \uparrow 2\beta$. Theorem 4 reaches the endpoint by a different argument, but retains eventual nondecrease of the parameters. For general odd nonexpansive maps, arbitrary oscillating schedules with $\limsup \lambda_n = 1$ and only $\sum_n \lambda_n(1-\lambda_n) = \infty$ are not treated here. Thus the unrestricted question in [1] is not resolved by this note. The Cauchy estimate uses the remaining orbit energy; it is not a uniform rate of strong convergence over all maps and initial states.

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